Unnamed_Unit

Unit 4 Variables

PDF Documents

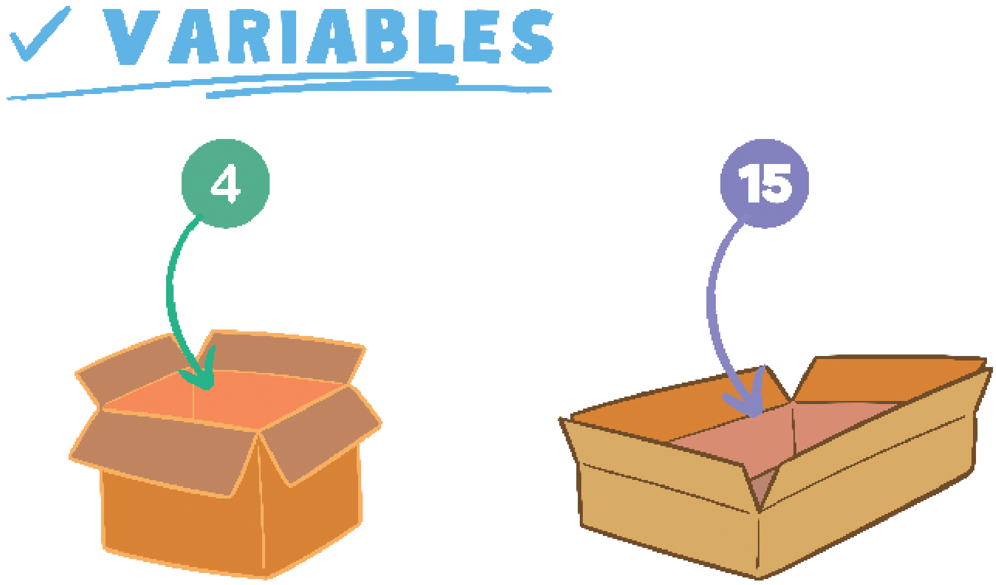
PDF Document 1: 17318421311137-Chapter 3_ unit4 Variables.pdf

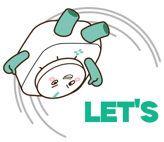
This PDF document is included in the unit materials.

UNIT 4 VARIABLES

Variables

Variables are like labeled boxes in a computer where we keep information we need, like numbers or names. They help us remember and change things in our code, making it smarter.

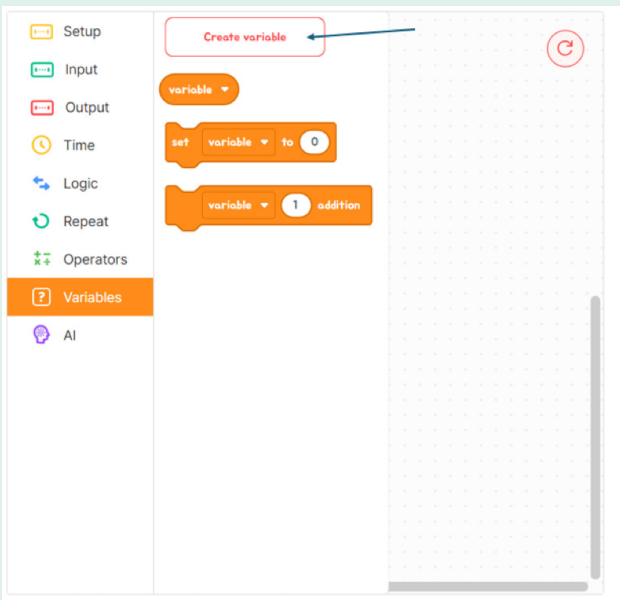


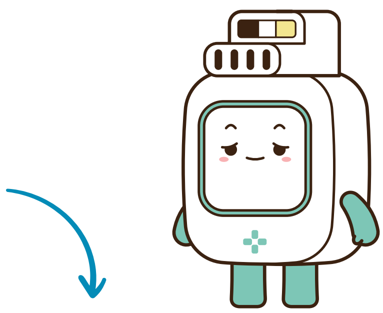


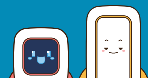
LET'S MAKE A VARIABLE IN CODE EDITOR:

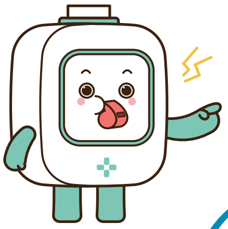
Step1

Go to the Variables section and click "Create Variable."



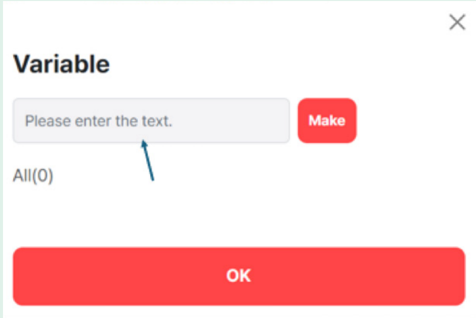






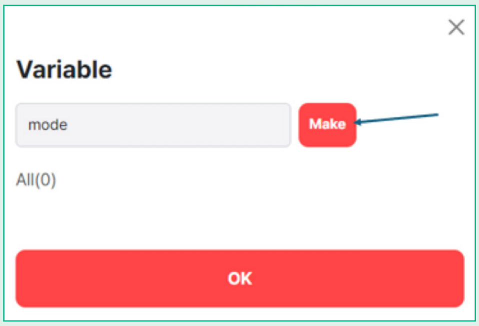
Step2

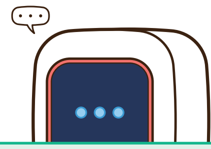
Type the name for your variable (like "mode").



Step3

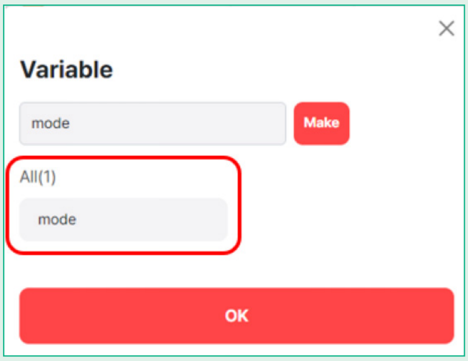
After typing the name, click "Make" to create it.





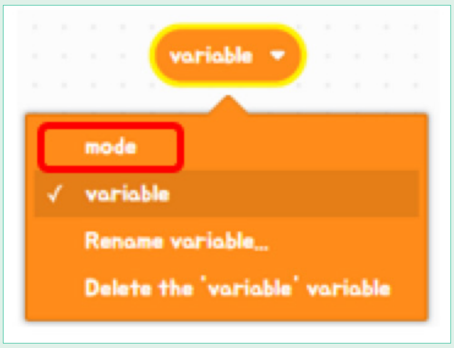
Step4

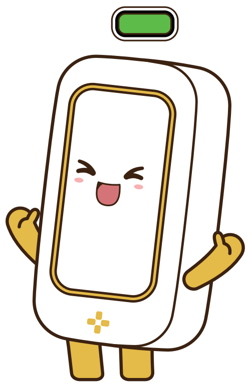
You'll then see your new variable in the list.



Step5

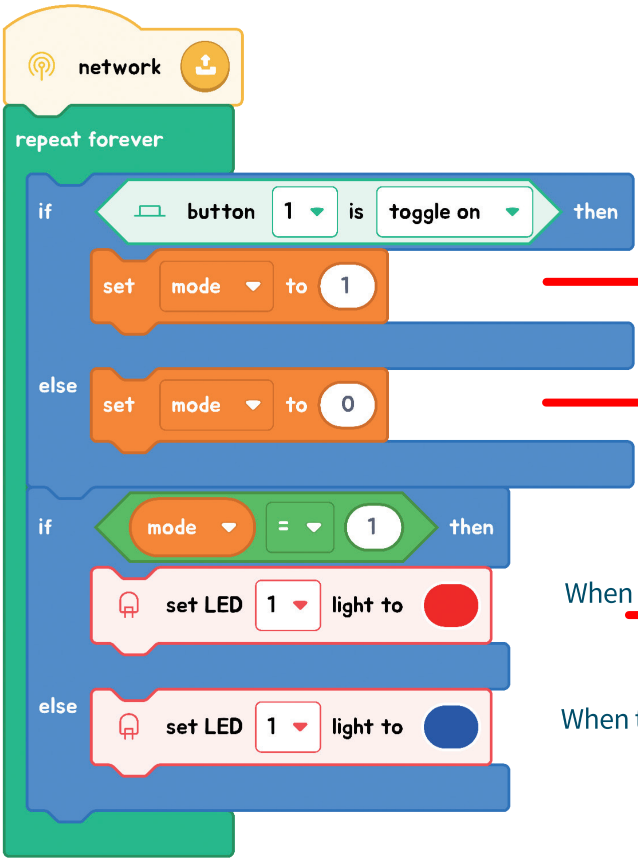
Click on the area where variables are shown to see the one you made.





EXPLANATION WITH CODE EXAMPLE

In our code, we're using a button and an LED light along with a variable named 'mode'. Here's what happens in our program:



When the button is on, the variable 'mode' is set to 1

When the button is off, the variable 'mode' is set to 0

When the variable 'mode' is 1, the LED should turn red.set to 0

When the variable 'mode' is 0, the LED should turn blue.set to 0

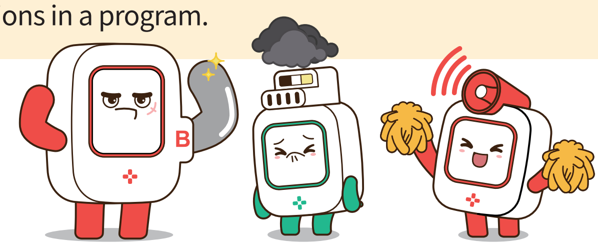
Code Insight

1. **Variable 'mode':** Think of 'mode' as the program's memory of whether the button is on or off. 'Mode' can be either 1 (on) or 0 (off).

2. **Checking the Button:** The first part of our code checks the button's state. If the button is on, 'mode' gets set to 1; if it's off, 'mode' goes to 0. This is how the program 'remembers' the button's state.

3. **Action Based on 'mode':**
   - **If 'mode' is 1:** The program sees 'mode' is 1 and knows the button was on, so it turns the LED red.
   - **If 'mode' is 0:** The program finds 'mode' is 0, it makes the LED blue.

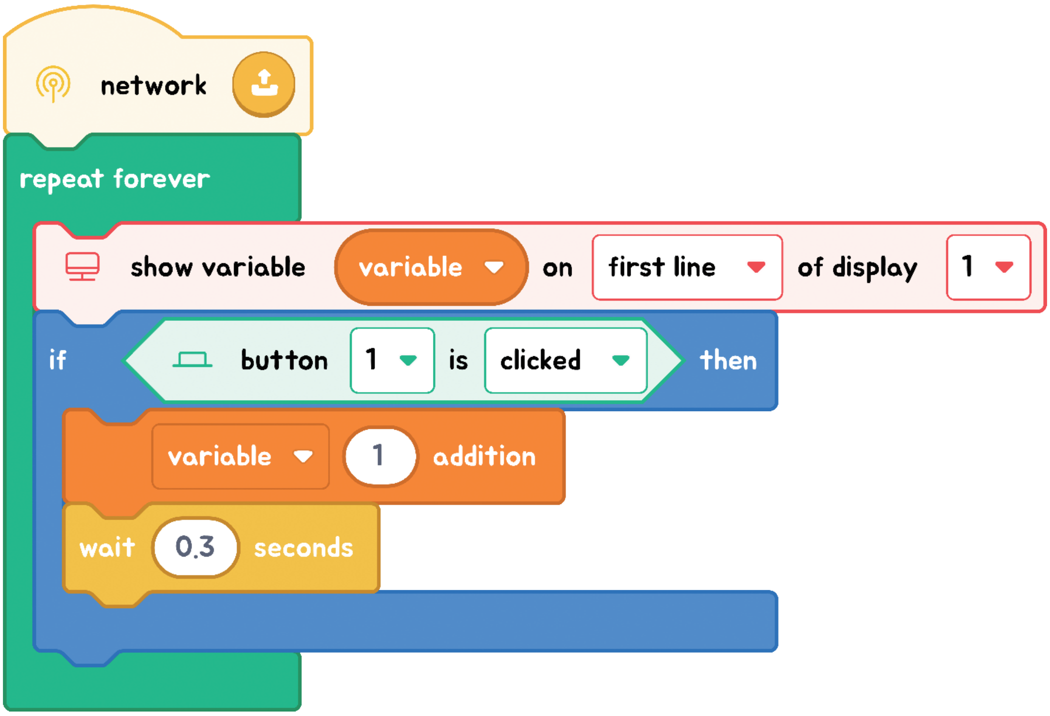
4. **The Role of 'mode':** The variable 'mode' acts as a switch for the LED's color, changing based on the button's state, showing how variables can control actions in a program.



Exercise

Exercise: Click Counter challenge

Every time you click the button, we want to add one more to our click counter (that's our variable). Using a 'plus' block in block coding, we can keep track of how many times the button has been clicked. When we first make a variable, it starts at 0, unless we tell it to start at a different number.



Code Insight

1. **Starting Point:** When we begin, our display screen is set to show a number(variable) called "clicks." Think of "clicks" like your score in a video game, which everyone can see on the screen.
2. **Click to Add:** Each time the button is pressed, we add 1 to our "clicks." Because the display is always showing "clicks," everyone can see the score go up each time the button is clicked.
3. **Updating the Score:** Just like how a scoreboard updates every time a player scores in a game, our display updates the "clicks" number every time the button is clicked, showing the new, higher score.
4. **To decrease the number with each button click:** switch to using a subtraction operation in the code.